

KILNDECK · REQUIREMENTS

KilnDeck requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	R2 and R4 restated, R12 and R13 added (KND-DDR-001); status on paper from KND-CAL-001 for the constructable design
0.1	2026-09-30	Amish Chadha	Initial scaffold

KilnDeck requirements

On paper, the constructable design meets ten of the thirteen requirements. R7 (floor temperature) is not met in full summer sun, R1 is met on the straight sides of the kiln but not on the curved ends, and R8 is met only against an interim bearing limit until the partner kiln is surveyed. R5 (dose repeatability) can only be judged by weighing doses at TRL 4. Every "met on paper" is a calculation in KND-CAL-001, not a test; verification is TRL 4 work.

Table 1. Requirements and their status on paper.

ID	Requirement	Target	Verification (TRL 4 or later)	Status on paper (KND-CAL-001)
R1	Fireman feeds every hole in a round without stepping onto the crust	0 steps onto the crust per feeding round	Observed trial over at least 20 feeding rounds	Met on the straight sides; not met on the curved ends , which this prototype does not serve
R2	Guardrail height and strength (restated)	Top rail 1.1 m (43 in) above the floor with a knee rail at 0.55 m and a 150 mm toe board; holds 1 kN (225 lbf) horizontal at the top rail, proof 1.5 kN without yield	CalRig proof-load test	Met: 81 to 100 MPa at 1 kN, 150 MPa at the proof load (yield 275 MPa)
R3	Deck rated load	250 kg (550 lb): fireman, full hopper and margin; proof at 1.5 times rated load	Proof load	Met: 18 MPa in the chords at the proof load; 0.5 mm deflection in use
R4	Force to move the loaded deck (restated)	200 N (45 lbf) or less at the hand wheel rim on level track	Force gauge on the rim	Met: about 64 N level, 128 N on a 2 % grade
R5	Fuel dose per pocket	Repeatable within 15 % of the set dose for dry coal	Weigh 20 doses	To verify at TRL 4; nominal 0.49 kg of coal per pocket
R6	Hopper capacity	At least one full feeding round for one fireman	Measured against the partner kiln's fuel use	Met: 52 L, about 42 kg of coal against about 30 kg a round (1.4 rounds)
R7	Floor temperature at the standing area	50 °C (122 °F) or less after 6 h on an operating kiln	Contact thermometer log during a firing shift	Not met in full sun: about 74 °C at 45 °C air (62 °C at 30 °C air); met at night (34 °C) and in shade (48 °C). The bare crust is about 150 °C
R8	Load on the kiln structure	Below a per-support limit set from a structural survey of the partner kiln	Load cell reading on site	Met against the interim limit of 0.20 MPa (0.15 MPa under a pad); cannot be confirmed until the survey
R9	Spout alignment with the feed hole	Within 25 mm (1 in)	Measurement at 10 holes	Met: about 17 mm along the kiln, 10 mm across
R10	Parking brake	Holds the loaded deck on a 5 % grade with no creep	Test on inclined track	Met: lock holds 727 N against 484 N
R11	Cost	Value-engineering target: USD 3,500 for one deck and a demonstration length of track	Costed bill of materials	Estimated cost of the constructable design: USD 2,627 (USD 873 under the target)
R12	Piece mass (new)	Every piece carried up the kiln 40 kg (88 lb) or less	Weigh each piece	Met: heaviest is a truss module at 39.2 kg
R13	Track retention (new)	The deck cannot run off a track end or drop off a track	Push test against the stops; lift test of one wheel	Met by design: end stops, buffers and derailment guards

■ Assumptions

- Kiln walls can carry the track pads without damage; to be confirmed kiln by kiln by a structural survey.
- The trench is about 6.0 m wide, the wall tops are at least 0.6 m wide and level with the crust within 100 mm, and feed holes are on a grid of about 1.0 m (KND-CAL-001, section 1).
- Firemen and owners accept a deck if it does not slow feeding.
- The track is laid in sections on the straight sides and moved with the fire.